Swapnil Botu

botu@berkeley.edu | 916-577-0628 | Berkeley, CA | linkedin.com/in/swapnil-botu | github.com/swapnilbotu

EDUCATION

University of California, Berkeley - Data Science - GPA: 4.00 — **Berkeley, CA**
- *Statistical Modeling, Data Visualization, Causal Inference, Python Programming, PyTorch* — *Expected: May 2028*

Folsom Lake College - Artificial Intelligence and Machine Learning Certificate (16 unit) - GPA: 4.00 — **Folsom, CA**
- *Python, Data Science, Machine Learning, Computer Vision, Natural Language Processing, Tableau* — *December 2025*

Sierra College - Embedded Systems Certificate (26 unit) - GPA: 4.00 — **Rocklin, CA**
- *Discrete Structures, Embedded Programming, Robotics/Electronics, C/C++, Java, OOP* — *August 2025*

WORK EXPERIENCE

Machine Learning Engineering Intern — **Sacramento, CA**
Home Details — *August 2026 – Present*
- Improved solar/pool detection recall from 84% → 95% while maintaining 90%+ precision across 1.1M+ Sacramento-area properties.
- Built Python/SQL pipelines for 2M+ property records, cutting data aggregation time 80% and replacing manual Sheets workflows.
- Raised lead-scoring lift 32% using property and geographic features, reducing wasted marketing outreach 35% across Sacramento.
- Analyzed outreach data across 10K+ leads, identifying 6 key conversion drivers and improving campaign response rates by 18%.

AI Evaluation & Machine Learning Engineering Intern — **Berkeley, CA**
Safe Robots — *June 2026 – September 2026*
- Raised safety-standards table classifier accuracy (60% → 94.6%); replaced keyword checks with weighted regex and semantic rules.
- Deployed the classifier via FastAPI, added Logfire monitoring, and reduced average inference latency by 74% (420 ms → 110 ms).
- Benchmarked 7 rule-based and ML pipelines using TF-IDF, BGE-M3 embeddings, Logistic Regression, Linear SVM, and hybrid features; identified BGE-M3 + Logistic Regression as the strongest learned pipeline (81.1% accuracy, 0.836 macro-F1).
- Built a versioned 8-class dataset of 77 real tables and 160 leakage-safe augmented examples from ISO 26262 and MIL-STD-882E.

Lead Software Development Intern — **Roseville, CA**
LearnHaus — *January 2025 – June 2025*
- Built Chrome Extension that boosted LearnHaus' platform usage by 54%, utilized Python, JavaScript, HTML, and CSS.
- Developed a Python keyword-based PDF scraper to identify and download relevant files, reducing manual search time by 45%.
- Authored three, 4 page technical analyses (OpenAI, HumeAI, Mixpanel) with integration recommendations and cost/feature analysis.

Artificial Intelligence Engineering Intern — **San Francisco, CA**
Reality AI Lab — *April 2024 – May 2025*
- Increased EdTech platform file support from 3 → 16 formats (433% growth), adding new text, audio, and video file types.
- Built an AI chatbot, flashcard generator, and quiz engine (Python + LangChain); adopted by 200+ students and 50+ teachers.

PROJECTS

ML Model for Microfinance Loan Reliability (First-Authored Peer-Reviewed Paper at Vanderbilt) | *SciKit-Learn, Pandas, Seaborn*
- Published in *Vanderbilt's International Journal* (<10% acceptance rate), predicting borrower reliability without credit history.
- Developed Decision Tree and Random Forest ML models analyzing 2.6M+ microfinance loan records to assess client reliability.
- Improved model performance on our imbalanced dataset using SMOTE, GridSearchCV, and advanced hyperparameter tuning.
- Achieved 86% precision and 96% recall in classifying reliable vs. unreliable borrowers.

Career Match: AI-powered App for Personalized Tech Career Exploration | *Gemini API, CareerOneStop API, Jinja2, Bootstrap*
- Developed an interactive career quiz that recommends curated career profiles used by 1,500+ high school students in Sacramento.
- Built a Gemini-powered AI chatbot, gamified badges, and a volunteer finder offering real-world opportunities with tailored guidance.

LEADERSHIP EXPERIENCE

Coding/Cyber Security Club: *President* — *August 2023 – May 2026;* **Roseville, CA**
- Supervised 2-3 major coding projects per semester, ensuring milestones are met and providing individual guidance to teams.
- Organized 2+ major events per semester, including hackathons, competitions, and networking events.

Folsom Lake College Machine Learning Bootcamp: *Teaching Assistant* — *February 2025 – May 2025;* **Folsom, CA**
- Delivered real-time technical support to 60+ college students, resolving 95% of issues within 5 minutes, boosted satisfaction by 40%.
- Graded 35+ projects and created materials on hyperparameter tuning, bias-variance, and GANs, improving subsequent scores 20%.

CERTIFICATIONS

Business AI-Accelerated Software Development (Course 005) - *Hewlett Packard Enterprise (HPE)* — *July 2026*

Break Fixes, Problem Solving, Testing, Digital Forensics & Agile (Course 004) - *Hewlett Packard Enterprise (HPE)* — *March 2026*

Clients/Servers, APIs, Security, Siloed Teams & Role Management (Course 003) - *Hewlett Packard Enterprise (HPE)* — *March 2025*

Deep Learning Specialization - *DeepLearning.AI* — *January 2025*

Machine Learning Specialization - *Stanford University* — *January 2025*